

A Guide to Understanding CWIS in IMIS



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Abbreviations

The following abbreviations are used throughout this manual. Refer back to this section whenever an unfamiliar term appears.

Abbreviation	Full Form	Meaning
AIT	Asian Institute of Technology	Academic partner involved in developing the base IMIS platform
BIN	Building Identification Number	Unique identifier for each building in IMIS. Used to link records across all downloaded files
BOD	Biochemical Oxygen Demand	Laboratory measure of organic matter in treated effluent — one of three parameters in SF-1g and SF-9
BWSMSS	Building-level Widely Safe Managed Sanitation System	The IMIS classification rule determining whether a building toilet is safely managed
CT	Community Toilet	A shared toilet block serving a specific group of nearby households
CWIS	Citywide Inclusive Sanitation	A planning approach ensuring safe sanitation for all city residents, with special focus on low-income areas
EAWAG	Swiss Federal Institute of Aquatic Science and Technology	Research institution whose FSM Toolbox informs rates and methods used in CWIS
FS	Fecal Sludge	The semi-solid waste accumulating at the bottom of pits, septic tanks, and holding tanks over time
FSM	Fecal Sludge Management	The full chain of managing waste from non-sewered toilets — collection, transport, treatment, disposal
FSMIS	Fecal Sludge Management Information System	The IMIS module that tracks desludging services, applications, and sludge collection records
FSTP	Fecal Sludge Treatment Plant	A facility that receives and treats sludge pumped from tanks by desludging trucks
GIS	Geographic Information System	Software used to capture and analyze location data — used for SF-3e distance calculation in IMIS
GPS	Global Positioning System	Satellite technology recording precise location of buildings and facilities during surveys
IHHL	Individual Household Latrine	A private toilet facility within a single household
IMIS	Integrated Municipal Information System	The city-wide digital platform developed by ISPL that stores all building, sanitation, and service data
ISPL	Innovative Solutions Pvt. Ltd	The technology company that developed and maintains IMIS
LIC	Low Income Community	Households or areas registered as low-income in IMIS. Identified by Is Low Income House = Yes OR LIC Name is filled
LPCD	Liters Per Capita Per Day	Standard unit for daily per-person water consumption. The FS rates (270, 280) in CWIS Settings are in mLPCD
mLPCD	Milliliters Per Capita Per Day	Correct unit for the FS accumulation rates: 270 mLPCD (sealed) and 280 mLPCD (unlined pit)
NSS	Non-Sewered Sanitation	Any toilet system not connected to a piped sewer — includes septic tanks, pits, and holding tanks

Abbreviation	Full Form	Meaning
OSS	On-Site Sanitation	A system where waste is contained and treated at or near the building (synonymous with NSS)
PT	Public Toilet	A toilet facility open to the general public — visitors, commuters, and residents
TSS	Total Suspended Solids	Measure of undissolved particles in treated effluent — second of the three parameters
WHO	World Health Organization	International body whose FSM guidelines inform the FS accumulation rates used in IMIS
WW	Wastewater	Liquid waste from households flowing through a sewer or drain network
WWTP	Wastewater Treatment Plant	A facility that treats liquid sewage flowing in through the sewer network

Color Symbology

Throughout this manual, specific colors are used consistently to help you identify the type of element being referenced. Familiarize yourself with these before reading the indicator sections.

Sample	Element Type	What it means
Buildings Containments.csv	Excel / CSV file name	The name of the file to download from IMIS. Appears in the middle column of every data table, after "File:". Always blue and bold.
Population of Building	Column / field name	The exact header of a column in the downloaded file. Appears with → prefix in the middle column, and highlighted inline in the "How to Derive" column. Always green.
✓ Example	□ Example box	A worked calculation using real data from the ISPL dataset. Shows numbers you can verify by downloading the files and applying the formula.
□ Note	□ Information box (blue)	Supplementary context or a unit clarification that helps interpret the formula correctly.
□ Note	□ Note box	An important caution, a design assumption, or a methodological consideration to keep in mind.
□ Step	□ Step box	Sequential instructions — such as the download steps — where order matters.

How to Use This Manual

This manual is structured in three layers:

- Section 1 — Download Guide: shows which IMIS module to visit and which file to download before you begin any calculation.
- Sections 2–8 — Indicator reference: each indicator shows its definition, formula, example.

□ Each indicator's data table has three columns: (1) the data point name, (2) the Excel or CSV file name and the exact column headers to use from that file, (3) the step-by-step derivation logic to apply.

1. Downloading Data from IMIS

Before calculating any indicator, download the required data files from IMIS. All files are available as Excel or CSV exports from the relevant module. The table below shows exactly where to go in IMIS for each file.

□ How to download from IMIS

Step 1: Log in to your IMIS instance.

Step 2: Click the module name in the side navigation bar (e.g. Building IMS, Fecal Sludge IMS).

Step 3: Click the sub-module from the sidebar after expanding the module (e.g. Buildings, Containments).

Step 4: Click the “Export to CSV” button to download data in CSV format.

Step 5: Save the file. The file name matches the "File to Download" column below.

1.1 Module-wise File Download Reference

IMIS Module	Sub-Module	File to Download	Used In Indicators
Building IMS	Buildings	Buildings.csv	EQ-1, SF-1a, SF-1b, SF-2a, SF-2b, SF-3, SF-3c, SF-5, SF-6
Building IMS	Low Income Community	Low Income Community.csv	EQ-1, SF-2a, SF-2b, SF-2c (LIC filter reference)
Fecal Sludge IMS	Containments	Buildings Containments.csv	EQ-1, SF-1a, SF-1b, SF-1d, SF-1f, SF-2a, SF-2b, SF-3, SF-3e, SF-5, SF-6
Fecal Sludge IMS	Containments	Containments.csv	Same file as above — downloaded via Buildings sub-module
Fecal Sludge IMS	Treatment Plant IMS > Treatment Plants	Treatment Plants.csv	SF-1d, SF-1e, SF-1f, SF-1g
Fecal Sludge IMS	Treatment Plant IMS > Performance Efficiency Test	Performance Efficiency Test.csv	SF-1g, SF-9
Fecal Sludge IMS	Emptying Service IMS > Application	Applications.csv	SF-2b, SF-2c (linking LIC buildings to their applications)
Fecal Sludge IMS	Emptying Service IMS > Emptying	Emptyings.csv	SF-1b, SF-1c, SF-1e, SF-2b, SF-7
Fecal Sludge IMS	Emptying Service IMS > Sludge Collection	Sludge Collections.csv	SF-1c, SF-2c
PT/CT IMS	Public / Community Toilets	Community Toilets.csv	SF-3, SF-3b, SF-3c, SF-3e, SF-4a, SF-4b
PT/CT IMS	PT Users Log	PT Users Log.csv	SF-4d
Utility IMS	Sewer Network	Sewer Network.csv	BWSMSS safety check (C-3, C-12)
Utility IMS	Drain Network	Drain Network.csv	BWSMSS safety check (C-4, C-5)
Public Health ISS	Water Samples	Water Samples.csv	SF-9 (water quality monitoring)

1.2 Key Columns in Each File

The tables below list the columns you will use from each file. These column names appear exactly as labeled in the downloaded Excel/CSV headers.

Buildings.csv	Key columns: BIN Population of Building Population with Private Toilet Female Population Is Low Income House LIC Name Toilet Connection Functional Use of Building Use Category of Building Community Toilet Name Sewer Code Drain Code
Buildings Containments.csv	Key columns: BIN Population of Building Population with Private Toilet Female Population Is Low Income House LIC Name Toilet Connection Containment Code Containment Type Containment Volume (m ³) Emptied Status Community Toilet Name Functional Use of Building
Emptyings.csv	Key columns: Application ID House Number (= BIN) Sludge Volume (m ³) Service Provider Name Date
Sludge Collections.csv	Key columns: Application ID Treatment Plant Name Sludge Volume (m ³) Date
Applications.csv	Key columns: BIN House Number Emptying Status Sludge Collection Status Application Date Service Provider Name
Treatment Plants.csv	Key columns: Name Capacity Per Day (m ³) Status Treatment Plant Type
Performance Efficiency Test.csv	Key columns: Treatment Plant Sample Date BOD (mg/l) TSS (mg/l) Ecoli Remark
Community Toilets.csv	Key columns: Toilet Type Toilet Name BIN Containment Connected Sewer Code Drain Code Adherence with Universal Design Principles No. of Seats for Male Users No. of Seats for Female Users Total Number of Urinals
PT Users Log.csv	Key columns: Toilet ID Date No. of Male Users No. of Female Users
Water Samples.csv	Key columns: Sample Date Sample Location Water Coliform Test Result

2. Introduction

This manual describes how each of the 22 CWIS indicators is calculated from IMIS data. For every indicator it shows what data points are required, which Excel file contains them, which columns to use, and what calculation steps produce the final number.

2.1 What is CWIS?

Citywide Inclusive Sanitation (CWIS) ensures safe, equitable sanitation for all city residents, with particular focus on low-income communities. The 22 indicators in this manual cover Equity and Safety outcomes.

2.3 Overview of CWIS Indicators in IMIS

Code	Category	Indicator Name	What It Measures
EQ-1	EQUITY	Ratio of LIC Access to Total Population Access	Compares the share of low-income residents with a safe private toilet against the city-wide average. Produces a ratio from 0 to 1.0 (or above).
SF-1a	SAFETY	Percentage of Population with Access to Safe, Private, Individual Toilets	The percentage of the total city population living in a building with a private toilet connected to a safely managed waste system.
SF-1b	SAFETY	Percentage of On-site Sanitation that has been Desludged	The percentage of all registered tanks and pits in the city that were emptied by a desludging vehicle during the reporting year.
SF-1c	SAFETY	Percentage of Collected FS Disposed at a Treatment Plant or Designated Site	Of all fecal sludge collected from tanks by desludging trucks, the percentage confirmed received at a treatment plant rather than dumped illegally.
SF-1d	SAFETY	FS Treatment Capacity as % of Total FS Generated from NSS (excl. safely disposed in situ)	Compares the fecal sludge treatment plant's annual processing capacity against the estimated annual sludge production from all NSS households that need external treatment.
SF-1e	SAFETY	FS Treatment Capacity as % of Total FS Collected from NSS Connections	Compares FSTP annual capacity against the actual volume of sludge that trucks brought to the plant during the year.
SF-1f	SAFETY	Wastewater Treatment Capacity as % of Total Wastewater Generated	Compares the wastewater treatment plant's annual capacity against the total wastewater the city generates, estimated from population and standard per-capita rates.
SF-1g	SAFETY	Effectiveness of FS/WW Treatment in Meeting Prescribed Standards for Effluent Discharge	The proportion of operational treatment plants whose effluent meets all three prescribed discharge standards simultaneously: BOD (organic matter), TSS (particles), and E.coli (bacteria).
SF-2a	SAFETY	Percentage of LIC Population with Access to Safe Individual Toilets	The percentage of residents in officially registered low-income areas who have a safe private toilet at home.

Code	Category	Indicator Name	What It Measures
SF-2b	SAFETY	Percentage of LIC, NSS, IHHLs that have been Desludged	The percentage of tanks and pits in low-income buildings that were emptied during the year.
SF-2c	SAFETY	Percentage of Collected FS from LIC Disposed at Treatment Plant or Designated Site	Of sludge collected from low-income areas, the percentage confirmed received at a treatment plant rather than dumped informally.
SF-3	SAFETY	Percentage of Dependent Population with Access to Safe Shared Facilities (CT/PT)	Of all city residents who do not have a private toilet at home (dependent population), the percentage who can access a safely managed community toilet or public toilet.
SF-3b	SAFETY	Percentage of CTs that Adhere to Principles of Universal Design	The percentage of community toilets that meet inclusive design standards: ramp access, grab rails, adequate door width, and a dedicated accessible cubicle.
SF-3c	SAFETY	Percentage of Users of CTs that are Women	The female share of the population linked to community toilets, measured from the gender composition of CT-dependent households.
SF-3e	SAFETY	Average Distance from the House to the Closest CT (meters)	The average straight-line distance in meters between each CT-dependent building and the community toilet it is assigned to.
SF-4a	SAFETY	Percentage of PTs where FS/WW Generated is Safely Transported to TP or Safely Disposed In Situ	The percentage of registered public toilets whose waste goes to a safely managed system — confirmed via their containment type or sewer connection.
SF-4b	SAFETY	Percentage of PTs that Adhere to Principles of Universal Design	The percentage of public toilets that meet universal design standards for accessibility.
SF-4d	SAFETY	Percentage of Users of PTs that are Women	The female share of actual public toilet users, recorded from daily tally counts by PT caretakers.
SF-5	SAFETY	Percentage of Educational Institutions where FS/WW is Safely Transported to TP or Safely Disposed In Situ	The percentage of all schools, colleges, and universities in the city with a safely managed toilet system.
SF-6	SAFETY	Percentage of Healthcare Facilities where FS/WW is Safely Transported to TP or Safely Disposed In Situ	The percentage of all hospitals, clinics, and health posts with a safely managed waste system.
SF-7	SAFETY	Percentage of Desludging Services Completed Mechanically or Semi-Mechanically (by a Gulper)	The percentage of all tank-emptying services completed using a mechanical vacuum truck or semi-mechanical pump rather than manual (hand) methods.
SF-9	SAFETY	Percentage of Tests in Compliance with Water Quality Standards for Fecal Coliform	The percentage of laboratory tests on treatment plant effluent that passed all three prescribed discharge standards (BOD, TSS, and E.coli) simultaneously.

2.3 The BWSMSS Safety Classification

Most indicators require classifying each building as safely managed (BWSMSS = True) or not. In the Buildings Containments.csv file, look at the Toilet Connection and Containment Type columns. A building is safely managed if its Containment Type is one of:

Containment Type value in Buildings Containments.csv	Why it is safe
Septic Tank connected to Soak Pit	Sealed tank; liquid filtered through soak pit underground
Lined Pit connected to a Soak Pit	Sealed pit; liquid filtered through soak pit underground
Double Pit	Alternating pits; waste decomposes safely on-site
Composting Toilet / UDDT / Ecosan	Full on-site treatment, no runoff
Onsite Treatment (Biogas, DEWATS)	Engineered on-site treatment system
Septic Tank connected to Sewer Network	Sewage treated at WWTP (check Sewer Code is filled)
Lined Pit connected to Sewer Network	Sewage treated at WWTP (check Sewer Code is filled)
Sewer Network	Building directly on sewer; sewage goes to WWTP

Any other Containment Type value — unlined pit, no outlet, open ground, water body — is NOT safely managed.

2.4 CWIS Settings — Rate Constants

Setting	Default	Used In
FS accumulation — sealed/lined containments (mL/person/day)	270	SF-1d (Part 1)
FS accumulation — unlined/permeable pits (mL/person/day)	280	SF-1d (Part 2)
Average water consumption (L/person/day)	150	SF-1f
Wastewater conversion factor	80%	SF-1f
Greywater factor — sewer-connected buildings	80%	SF-1f
Greywater factor — non-sewer buildings	80%	SF-1f

3. Equity Indicator

EQ-1: Ratio of LIC Access to Total Population Access

Outcome	EQUITY — Access to safely managed sanitation
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Definition

Measures whether low-income residents have proportionally equal access to safe private sanitation compared to the city as a whole. A value of 1.0 means perfect equity; below 1.0 means LIC residents have less access.

Formula

$$EQ-1 = (LIC \text{ safe-toilet rate}) / (City \text{ safe-toilet rate})$$

$$LIC \text{ safe-toilet rate} = a / b \quad \quad \quad City \text{ safe-toilet rate} = c / d$$

Var	Meaning	Definition
a	LIC safe pop	Population with Private Toilet for buildings that are LIC AND safely managed
b	LIC total pop	Population of Building for all LIC buildings
c	City safe pop	Population with Private Toilet for all safely managed buildings
d	City total pop	Population of Building for all buildings

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
a — LIC safe pop	File: Buildings Containments.csv → Population with Private Toilet → Is Low Income House → LIC Name → Containment Type → Toilet Connection	Filter rows where Is Low Income House = "Yes" OR LIC Name is filled AND Containment Type is safe (see Section 2.2). SUM Population with Private Toilet for matching rows.

Data Point	Excel / CSV File Columns Used	How to Derive
b — LIC total pop	File: Buildings Containments.csv → Population of Building → Is Low Income House → LIC Name	Filter rows where Is Low Income House = "Yes" OR LIC Name is filled. SUM Population of Building.
c — City safe pop	File: Buildings Containments.csv → Population with Private Toilet → Containment Type → Toilet Connection	Filter rows where Containment Type is safe (no LIC filter). SUM Population with Private Toilet.
d — City total pop	File: Buildings.csv → Population of Building	SUM Population of Building for all rows. No filter.

Example



Low-income residents with a safe private toilet = 17

Total low-income population = 222

LIC safe-toilet rate = $17 \div 222 = 0.077$ (7.7%)

City residents with a safe private toilet = 59,164

Total city population = 101,254

City safe-toilet rate = $59,164 \div 101,254 = 0.584$ (58.4%)

EQ-1 = $0.077 \div 0.584 = 0.131$

This means low-income residents have only about 13% of the safe-toilet access enjoyed by the average city resident — a significant gap that calls for targeted investment.

4. Safety Indicators — City-wide

SF-1a: Percentage of Population with Access to Safe Individual Toilets

Outcome	SAFETY — Safe individual sanitation access
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Definition

The share of the total city population that lives in a building with a private toilet and a safely managed sanitation system.

Formula

$$SF-1a = \text{Population with safe private toilet} / \text{Total city population} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Numerator (safe pop with private toilet)	File: Buildings Containments.csv → Population with Private Toilet → Containment Type → Toilet Connection	Filter rows where Containment Type is safe (see Section 2.2). SUM Population with Private Toilet.
Denominator (total city population)	File: Buildings.csv → Population of Building	SUM Population of Building for all rows.

Example



Residents with a safe private toilet = 59,164

Total city population = 101,254

$$SF-1a = 59,164 \div 101,254 \times 100 = 58.4\%$$

About 58 out of every 100 residents have a safely managed private toilet at home. The remaining 42% — over 42,000 people — still lack safe private sanitation.

SF-1b: Percentage of On-site Sanitation that has been Desludged

Outcome SAFETY — Desludging coverage

Definition

The proportion of all registered NSS containments that were emptied during the year, measured by counting annual service records against the total registered containment stock.

Formula

SF-1b = Number of emptying services in the year / Total registered tanks and pits × 100

□ Use COUNT of rows in Emptyings.csv (each row = one completed service), NOT the Emptied Status column in Buildings Containments.csv. The Emptied Status flag accumulates over all years and would overcount.

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Numerator (services this year)	File: Emptyings.csv → Application ID (count rows) → Date (filter to reporting year)	Filter rows where Date falls within the reporting year. COUNT rows — each row is one completed service. Each record represents one completed tank or pit emptying.
Denominator (total registered tanks/pits)	File: Buildings Containments.csv → Containment Type	COUNT the number of rows with the Containment Type of “Septic Tank” or “Pit/ Holding Tank” or any NSS connection.

Example

□ Emptying service records in the year = 1,376
Containment with NSS connection = 11,752

SF-1b = $1,376 \div 11,752 \times 100 = 11.7\% \approx 12\%$
Only 1 in 8 tanks was emptied during the year.

SF-1c: Percentage of Collected FS Disposed at a Treatment Plant or Designated Disposal Site

Outcome SAFETY — Safe FS disposal

Definition

Of all fecal sludge collected from tanks by desludging trucks, what percentage was confirmed received at the treatment plant rather than being dumped illegally.

Formula

$$\text{SF-1c} = \frac{\text{Volume confirmed at treatment plant}}{\text{Total volume collected}} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Numerator (volume to treatment plant)	File: Sludge Collections.csv → Sludge Volume (m ³) → Application ID	SUM Sludge Volume (m³) for all rows. Each row = sludge confirmed received at the plant. Link to Emptyings via Application ID for filtering a specific year.
Denominator (total volume collected)	File: Emptyings.csv → Sludge Volume (m ³) → Date	Filter Emptyings.csv by Date to the reporting year. SUM Sludge Volume (m³) .

Example



Volume confirmed at plant = 2,055.22 m³

Total collected = 2,055.22 m³

SF-1c = $2,055.22 \div 2,055.22 \times 100 = 100\%$

All collected sludge reached the treatment plant — no illegal dumping recorded.

SF-1d: FS Treatment Capacity as % of Total FS Generated from NSS (excl. safely disposed in situ)

Outcome SAFETY — Treatment capacity adequacy

Definition

Compares FSTP annual capacity against the estimated sludge generated by NSS households whose waste must be trucked to the FSTP. Soak pit and double pit buildings are excluded because they treat waste safely on-site. A result above 100% means the FSTP has enough capacity.

Formula

$$\text{SF-1d} = \text{FSTP annual capacity (m}^3\text{/yr)} / \text{Total FS generated (m}^3\text{/yr)} \times 100$$

$$\text{FS generated} = (\text{pop_sealed} \times 270 \text{ mL} \times 365 / 1,000,000) + (\text{pop_unlined} \times 280 \text{ mL} \times 365 / 1,000,000)$$

□ 270 and 280 are in milliliters per person per day. Dividing by 1,000,000 converts mL to m³ so units match the capacity column (m³/day) in Treatment Plants.

Var	Meaning	Definition
FSTP cap	FSTP annual capacity	Sum Capacity Per Day (m ³) × 365 for FSTP rows where Status = Operational
pop_sealed	Sealed containment population	Population in buildings with sealed/lined containments NOT on the sewer AND NOT in-situ safe
pop_unlined	Unlined pit population	Population in buildings whose Containment Type is an unlined or permeable pit
270 mL	FS rate — sealed	Standard accumulation rate: 270 mL/person/day = 98.6 L/person/year
280 mL	FS rate — unlined	Slightly higher: soil ingress adds bulk → 280 mL/person/day = 102.2 L/person/year

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
FSTP annual capacity	File: Treatment Plants.csv → Capacity Per Day (m ³) → Status → Treatment Plant Type	Filter where Treatment Plant Type = "FSTP" or "Co-treatment" AND Status = "Operational". SUM Capacity Per Day (m ³) × 365.

Data Point	Excel / CSV File Columns Used	How to Derive
pop_sealed	File: Buildings Containments.csv → Population of Building → Containment Type → Toilet Connection	Filter rows where Containment Type contains "Septic Tank" or "Lined Pit" AND does NOT contain "Soak Pit" or "Sewer Network" AND Toilet Connection ≠ "Sewer Network". SUM Population of Building .
pop_unlined	File: Buildings Containments.csv → Population of Building → Containment Type	Filter rows where Containment Type contains "Unlined" or "Permeable" or is a basic open pit. SUM Population of Building .
FS generated (derived)	Derived from above	$(\text{pop_sealed} \times 270 / 1,000,000 \times 365) + (\text{pop_unlined} \times 280 / 1,000,000 \times 365)$. Result in m ³ /year.

Example



Treatment Plants.csv: FSTP, Operational, Capacity = 20 m³/day → $20 \times 365 = 7,300$ m³/year

pop_sealed = 10,944 → $10,944 \times 270 / 1,000,000 \times 365 = 1,079$ m³/year

pop_unlined = 1,031 → $1,031 \times 280 / 1,000,000 \times 365 = 105$ m³/year

Total FS generated = $1,079 + 105 = 1,184$ m³/year

SF-1d = $7,300 \div 1,184 \times 100 = 617\%$

The FSTP has more than 6× the needed capacity — desludging outreach needs to scale up.

SF-1e: FS Treatment Capacity as % of Total FS Collected from NSS Connections

Outcome	SAFETY — Treatment vs collected FS
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Definition

Compares FSTP capacity against the actual sludge volume received from trucks this year. A very high result means the plant is underused. Unlike SF-1d (which uses estimated generation), SF-1e uses the actual collected volume.

Formula

$$\text{SF-1e} = \text{FSTP annual capacity} / \text{Total sludge volume collected this year} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
FSTP annual capacity	File: Treatment Plants.csv → Capacity Per Day (m ³) → Status → Treatment Plant Type	Same as SF-1d. Filter FSTP rows, Status = Operational. SUM Capacity Per Day × 365.
Total sludge collected	File: Emptyings.csv → Sludge Volume (m ³) → Date	Filter Emptyings.csv by Date to the reporting year. SUM Sludge Volume (m ³).

Example

□ **FSTP capacity = 7,300 m³/year**

Total sludge from Emptyings.csv (1,376 rows) = 2,055 m³

$$\text{SF-1e} = 7,300 \div 2,055 \times 100 = 355\%$$

The plant ran at about 28% of capacity — room for significantly more services.

SF-1f: Wastewater Treatment Capacity as % of Total Wastewater Generated from Sewered Connections and Greywater Generated from non-Sewered Connections

Outcome

SAFETY — WW treatment coverage

Definition

Compares WWTP annual capacity against the total wastewater the city generates, estimated from population and standard per-capita rates. The denominator covers all four sanitation groups — sewer-connected, onsite-to-sewer, onsite-not-sewer, and pit/open — summed across the entire city population.

Formula

$$\text{SF-1f} = \text{WWTP annual capacity} / \text{Total wastewater generated} \times 100$$

$$\text{Total WW} = \text{Total city population} \times (150 \text{ L/p/day} \div 1,000) \times 80\% \times 365$$

□ 150 L/p/day is the standard urban water consumption figure in LITERS. Divide by 1,000 (not 1,000,000) to convert to m³. The 80% factor represents the share of consumed water that becomes wastewater.

Var	Meaning	Definition
pop_s1	Sewer network	Buildings where Toilet Connection = "Sewer Network"
pop_s2	Onsite → sewer	Buildings with a tank/pit whose Containment Type includes "Sewer Network" as the outlet
pop_s3	Onsite, not sewer	Buildings with a tank/pit connected to soak pit, drain, no outlet, water body
pop_s4	Pit / open	Buildings using unlined pits or discharging to open ground
150 L/p/day	Water use rate	Standard LPCD — divide by 1,000 to convert liters to m ³
80%	WW conversion	Of water used, 80% becomes wastewater entering the system

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
WWTP annual capacity	File: Treatment Plants.csv → Capacity Per Day (m ³) → Status → Treatment Plant Type	Filter where Treatment Plant Type = "Centralized WWTP" or "WWTP" AND Status = "Operational". SUM Capacity Per Day × 365.
City population (all 4 groups)	File: Buildings Containments.csv → Population of Building → Toilet Connection → Containment Type	Group buildings by sanitation type using Toilet Connection and Containment Type . SUM Population of Building for all groups combined. For SF-1f the total is the whole city population.
Total WW (derived)	Derived from above	Total city pop × (150 ÷ 1,000) × 0.80 × 365. Result in m ³ /year. If city has a WWTP, compute this; otherwise indicator is NaN.

Example



WWTP, Operational, Capacity = 30 m³/day → 30 × 365 = 10,950 m³/year

Total city population from Buildings Containments.csv = 101,049

Total WW = 101,049 × (150 ÷ 1,000) × 0.80 × 365 = 4,425,946 m³/year

SF-1f = 10,950 ÷ 4,425,946 × 100 = 0.25%

The small 30 m³/day WWTP serves about 250 people's worth of wastewater in a largely NSS city.

SF-1g: Effectiveness of FS/WW Treatment in Meeting Prescribed Standards for Effluent Discharge and Biosolids Disposal

Outcome

SAFETY — Treatment effectiveness

Definition

The proportion of operational treatment plants whose effluent meets all three quality standards simultaneously: BOD (organic matter), TSS (particles), and E.coli (bacteria). A plant fails if any one is above its threshold.

Formula

$$SF-1g = \text{Plants passing all three tests} / \text{Total operational plants} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Lab test results per plant	File: Performance Efficiency Test.csv → Treatment Plant → BOD (mg/l) → TSS (mg/l) → Ecoli → Sample Date	For each plant name. Check: BOD ≤ prescribed limit AND TSS ≤ prescribed limit AND Ecoli ≤ prescribed limit. Mark each plant as Pass or Fail. Note: The three threshold values (BOD standard, TSS standard, E.coli standard) are set in IMIS Settings → CWIS → Treatment Plant Performance.
Total operational plants	File: Treatment Plants.csv → Status	COUNT rows where Status = "Operational".

Example

- Performance Efficiency Test.csv has 2 plant entries.

FSTP row: BOD=45, TSS=70, Ecoli=350 — all within thresholds → PASS

WWTP row: Ecoli=650 — exceeds threshold → FAIL

Passing plants = 1 | Total operational = 2

$$SF-1g = 1 \div 2 \times 100 = 50\%$$

5. Safety Indicators — LIC-specific

SF-2a: Percentage of LIC Population with Access to Safe Individual Toilets

Outcome

SAFETY — LIC safe sanitation access

Definition

The LIC numerator rate (a/b) from EQ-1 expressed as a percentage. Shows what share of low-income residents have a safe private toilet.

Formula

$$\text{SF-2a} = \frac{\text{LIC population with safe private toilet}}{\text{Total LIC population}} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Numerator	File: Buildings Containments.csv → Population with Private Toilet → Is Low Income House → LIC Name → Containment Type → Toilet Connection	Filter: Is Low Income House = "Yes" OR LIC Name is filled AND Containment Type is safe. SUM Population with Private Toilet.
Denominator	File: Buildings Containments.csv → Population of Building → Is Low Income House → LIC Name	Filter: Is Low Income House = "Yes" OR LIC Name is filled (no safety filter). SUM Population of Building.

Example



LIC safe pop = 17

Total LIC pop = 222

SF-2a = $17 \div 222 \times 100 = 7.7\% \approx 8\%$ (vs. city-wide 61%)

SF-2b: Percentage of LIC, NSS, IHHLs that have been Desludged

Outcome

SAFETY — LIC desludging coverage

Definition

The proportion of tanks and pits in LIC buildings that were emptied during the year.

Formula

$$\text{SF-2b} = \frac{\text{Emptying services in LIC buildings}}{\text{Total LIC tanks and pits}} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
LIC building BINs	File: Buildings Containments.csv → BIN → Is Low Income House	Filter: Is Low Income House = "Yes". Collect the list of BIN values — this is your LIC BIN set.
Numerator (LIC services this year)	File: Applications.csv → BIN → Application Date File: Emptyings.csv → Application ID → Date	From Applications.csv, filter rows where BIN is in the LIC BIN set. Get their Application IDs. From Emptyings.csv, COUNT rows for those Application IDs where Date is in the reporting year.
Denominator (LIC tanks)	File: Buildings Containments.csv → Containment Code → BIN	Filter rows where BIN is in the LIC BIN set. COUNT Containment Type with values "Septic Tank", "Pit/ Holding Tank" or any NSS connection

Example



LIC Buildings identified: 55 buildings
Containment with Septic Tank or Pit/ Holding Tank for those Buildings = 27
Emptyings records linked to LIC applications in the year = 5

$$\text{SF-2b} = 5 \div 27 \times 100 = 18.5\% \approx 19\%$$

SF-2c: Percentage of Collected FS (collected from LIC) Disposed at Treatment Plant or Designated Site

Outcome

SAFETY — LIC safe FS disposal

Definition

Of sludge collected specifically from LIC buildings, what share reached the treatment plant.

Formula

$$\text{SF-2c} = \frac{\text{LIC sludge volume confirmed at treatment plant}}{\text{LIC sludge volume collected}} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
LIC application IDs	File: Applications.csv → Application Date → BIN	Filter Applications.csv where BIN is in the LIC BIN set AND Application Date is in the reporting year. Collect the Application ID values.
Numerator	File: Sludge Collections.csv → Application ID → Sludge Volume (m³)	Filter Sludge Collections.csv where Application ID is in the LIC application ID set. SUM Sludge Volume (m³).
Denominator	File: Emptyings.csv → Application ID → Sludge Volume (m³)	Filter Emptyings.csv where Application ID is in the LIC application ID set. SUM Sludge Volume (m³).

Example

□

2 LIC application IDs identified.

Sludge Collections volume for those IDs = 2 m³

Emptyings volume for those IDs = 2 m³

SF-2c = $2 \div 2 \times 100 = 100\%$ (if all sludge reached the plant)

6. Safety Indicators — Community Toilets (CT)

SF-3: Percentage of Dependent Population with Access to Safe Shared Facilities (CT/PT)

Outcome

SAFETY — Shared facility access

Definition

Of all residents without a private toilet at home (dependent population), what share can reach a safely managed community toilet or public toilet.

Formula

$$\text{SF-3} = (\text{Population using safe CTs} + \text{Population covered by safe PTs}) / \text{Dependent population} \times 100$$

Dependent population = Total city population - Population with a private toilet

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Dependent population	File: Buildings.csv → Population of Building → Population with Private Toilet	SUM Population of Building for all rows. Subtract SUM Population with Private Toilet for all rows.
CT user population	File: Buildings.csv → Community Toilet Name → Population of Building	Filter Buildings.csv where Community Toilet Name is filled. These are CT-dependent buildings. SUM Population of Building.
CT safety status	File: Community Toilets.csv → Toilet Name → Containment Connected → Sewer Code → Toilet Type	For each CT (Toilet Type = "Community Toilet"), check whether Containment Connected links to a safe containment type. Mark each CT as safe or not.
Safe CT user population	File: Buildings.csv → Community Toilet Name → Population of Building	Filter Buildings.csv to buildings whose Community Toilet Name is in the safe CT name list. SUM Population of Building.
PT safety & coverage	File: Community Toilets.csv → Toilet Type = "Public Toilet" → Containment Connected	Apply same safety check for public toilets. PT coverage is estimated from the ward-level dependent population in wards served by safe PTs.

Data Point	Excel / CSV File Columns Used	How to Derive
	→ Sewer Code	

Example

□

Dependent pop = 101,238 – 71,542 = 29,696

Safe CT: 1 of 2 CTs is safe → Buildings with that CT name: population = 135

Safe PT: all 3 PTs safe → ward-level dep pop in those wards = 11,797

SF-3 = $(135 + 11,797) \div 29,696 \times 100 \approx 40\%$

SF-3b: Percentage of CTs that Adhere to Principles of Universal Design

Outcome

SAFETY — CT inclusivity

Formula

SF-3b = $\text{CTs meeting universal design} / \text{Total CTs} \times 100$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Numerator	File: Community Toilets.csv → Toilet Type → Adherence with Universal Design Principles	Filter where Toilet Type = "Community Toilet" AND Adherence with Universal Design Principles = "Yes". COUNT rows.
Denominator	File: Community Toilets.csv → Toilet Type	COUNT all rows where Toilet Type = "Community Toilet".

Example

□

Community Toilets.csv: 2 CT rows. 1 has Adherence = "Yes", 1 has "No".

SF-3b = $1 \div 2 \times 100 = 50\%$

SF-3c: Percentage of Users of CTs that are Women

Outcome

SAFETY — CT gender equity

Definition

The female share of residents in buildings that are linked to community toilets, computed from the building survey gender data rather than seat counts.

Formula

$$\text{SF-3c} = \frac{\text{Female population in CT-linked buildings}}{\text{Total population in CT-linked buildings}} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
CT-linked buildings	File: Buildings.csv → Community Toilet Name → BIN	Filter Buildings.csv where Community Toilet Name is filled AND the name appears in the Community Toilets.csv Toilet Name column (Toilet Type = "Community Toilet"). These are CT-dependent households.
Numerator	File: Buildings.csv → Female Population	SUM Female Population for CT-linked buildings.
Denominator	File: Buildings.csv → Population of Building	SUM Population of Building for CT-linked buildings.

Example



CT-linked buildings: Female Population = 73
Population of Building = 205
SF-3c = $73 \div 205 \times 100 = 35.6\% \approx 36\%$

SF-3e: Average Distance from the House to the Closest CT (meters)

Outcome

SAFETY — CT proximity

Definition

The average straight-line distance between each CT-dependent building and its assigned community toilet, computed from GPS coordinates stored in IMIS using PostGIS. This is calculated automatically by the IMIS system — it does not require a manual GIS step.

Formula

SF-3e = Average of straight-line distances (GPS coordinates) between CT-dependent buildings and their assigned CT (meters)

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Building GPS coordinates	File: Buildings.shp → BIN → Community Toilet Name (GPS coordinates stored in IMIS — not exported in CSV)	The spatial coordinates of buildings are stored in the IMIS database. The distance calculation is run automatically when the CWIS indicator is generated.
CT GPS coordinates	File: Community Toilets.shp → Toilet Name → Toilet Type (GPS coordinates in IMIS database)	Same as above — CT coordinates are in the IMIS database. The system calculates straight-line distance between each building and its assigned CT using the Community Toilet Name link.

Example



48 buildings linked to CTs via Community Toilet Name column.
IMIS computes straight-line distance for each pair using GPS coordinates.
Average of 48 distances = 75 meters
SF-3e = 75 m

7. Safety Indicators — Public Toilets (PT)

SF-4a: Percentage of PTs where FS/WW is Safely Transported to Treatment Plants or Disposed In-Situ

Outcome

SAFETY — PT safe disposal

Formula

$$\text{SF-4a} = \frac{\text{Public toilets with safe waste system}}{\text{Total public toilets}} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
PT safety status	File: Community Toilets.csv → Toilet Type → Containment Connected → Sewer Code → Drain Code	Filter where Toilet Type = "Public Toilet". For each PT: if Containment Connected links to a safe containment type (look up that code in "Buildings Containments.csv")
Numerator	Community Toilets.csv (filtered)	COUNT PT rows marked as safe.
Denominator	File: Community Toilets.csv → Toilet Type	COUNT all rows where Toilet Type = "Public Toilet".

Example



3 PT rows. All 3 have Containment Connected → Septic Tank → Soak Pit (safe).

$$\text{SF-4a} = 3 \div 3 \times 100 = 100\%$$

SF-4b: Percentage of PTs that Adhere to Principles of Universal Design

Outcome

SAFETY — PT inclusivity

Formula

$$\text{SF-4b} = \frac{\text{PTs meeting universal design}}{\text{Total PTs}} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Numerator	File: Community Toilets.csv → Toilet Type → Adherence with Universal Design Principles	Filter where Toilet Type = "Public Toilet" AND Adherence with Universal Design Principles = "Yes". COUNT rows.
Denominator	File: Community Toilets.csv → Toilet Type	COUNT all rows where Toilet Type = "Public Toilet".

Example



3 PT rows in Community Toilets.csv. 1 has Adherence = "Yes".
 $SF-4b = 1 \div 3 \times 100 = 33\%$

SF-4d: Percentage of Users of PTs that are Women

Outcome SAFETY — PT gender equity

Definition

The female share of actual PT users, recorded by PT caretakers via daily tally counts. This uses actual usage data — seat counts are NOT used.

Formula

$$SF-4d = \text{Total female PT users} / \text{Total PT users (male + female)} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Female users	File: PT Users Log.csv → No. of Female Users → Toilet ID → Date	Filter PT Users Log by Date to the reporting year and Toilet ID to public toilet entries. SUM No. of Female Users.
Total users	File: PT Users Log.csv → No. of Male Users → No. of Female Users	SUM (No. of Male Users + No. of Female Users) for the same filtered rows.

Example



PT Users Log.csv: Male users = 2,500 | Female users = 250 (full year, all PTs)
 $SF-4d = 250 \div (2,500 + 250) \times 100 = 9.1\% \approx 9\%$

8. Safety Indicators — Institutional Buildings

SF-5: Percentage of Educational Institutions where FS/WW is Safely Managed

Outcome SAFETY — Schools and colleges

Formula

$$SF-5 = \text{Safe educational buildings} / \text{Total educational buildings} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Educational buildings	File: Buildings Containments.csv → Functional Use of Building → BIN	Filter where Functional Use of Building = "Educational". COUNT unique BIN values.
Safe educational buildings	File: Buildings Containments.csv → Containment Type → Toilet Connection → BIN	From the educational BIN set, further filter where Containment Type is safe (see Section 2.2). COUNT unique BIN values.

Example



There are 96 Educational Buildings in the City
Of those, 84 have a safe Containment Type
SF-5 = $84 \div 96 \times 100 = 87.5\% \approx 88\%$

SF-6: Percentage of Healthcare Facilities where FS/WW is Safely Managed

Outcome SAFETY — Hospitals and clinics

Formula

$$SF-6 = \text{Safe healthcare buildings} / \text{Total healthcare buildings} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Healthcare buildings	File: Buildings Containments.csv → Functional Use of Building → BIN	Filter where Functional Use of Building = "Health Institution". COUNT unique BIN values.
Safe healthcare buildings	File: Buildings Containments.csv → Containment Type → Toilet Connection → BIN	From the healthcare BIN set, further filter where Containment Type is safe. COUNT unique BIN values.

Example



City has 13 Health Care Facilities Buildings
 All 13 have safe Containment Type
 $SF-6 = 13 \div 13 \times 100 = 100\%$

9. Safety Indicators — Service Delivery

SF-7: Percentage of Emptying Services Provided Mechanically or Semi-Mechanically

Outcome SAFETY — Mechanical desludging

Definition

The proportion of recorded desludging services done by a registered mechanical provider. In IMIS, every service logged through the formal system is assumed to use mechanical equipment.

Formula

$$SF-7 = \text{Services by registered providers} / \text{Total services} \times 100$$

☐ In IMIS design, every service in Emptyings.csv is assumed mechanical because only registered providers log services here. As a result this indicator will always show 100% from the data download. Field monitoring is needed to catch any unregistered manual services.

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Total services (= mechanical services)	File: Emptyings.csv → Application ID → Service Provider Name → Date	Filter by Date to the reporting year. COUNT all rows. Since all logged services are by registered providers, numerator = denominator = this count.

Example

☐ Emptyings.csv rows in the year = 1,376. All have a Service Provider Name filled.
 $SF-7 = 1,376 \div 1,376 \times 100 = 100\%$

SF-9: Percentage of Tests in Compliance with Water Quality Standards for Fecal Coliform

Outcome SAFETY — Effluent quality compliance

Definition

The proportion of laboratory tests on treatment plant output that passed all three prescribed standards simultaneously: BOD, TSS, and E.coli. A test fails if any one is above its threshold.

Formula

$$SF-9 = \text{Tests passing all three quality measurements} / \text{Total tests} \times 100$$

Data Points and Derivation Steps

Data Point	Excel / CSV File Columns Used	How to Derive
Test results	File: Performance Efficiency Test.csv → Treatment Plant → Sample Date → BOD (mg/l) → TSS (mg/l) → Ecoli	Filter by Sample Date to the reporting year. For each row, check: BOD ≤ threshold AND TSS ≤ threshold AND Ecoli ≤ threshold. Mark each row Pass or Fail.
Prescribed thresholds	<i>Source: CWIS Settings in IMIS</i>	Check the CWIS Settings screen in IMIS for the BOD, TSS, and E.coli discharge standard thresholds.
Numerator	Performance Efficiency Test.csv (filtered)	COUNT rows marked as Pass.
Denominator	Performance Efficiency Test.csv	COUNT all rows for the reporting year.

Example



Performance Efficiency Test.csv: 8 rows for the year.
5 rows: BOD, TSS, Ecoli all within thresholds → PASS
3 rows: Ecoli above threshold → FAIL

$$SF-9 = 5 \div 8 \times 100 = 62.5\% \approx 63\%$$